

Embedded Systems – Lecture 3

Arduino Programming: Review, Functions & Demonstrations

Lecture type: Theoretical with short code demonstrations

Learning objectives

By the end of this lecture, students should be able to explain the main Arduino functions introduced previously, distinguish built-in functions from user-defined functions, understand the flow of a simple Arduino program, and predict the output of short code examples.

1. Quick Review of Arduino

Arduino is a microcontroller-based platform used to build embedded-system prototypes.

The basic Arduino program contains two important functions: `setup()` and `loop()`.

`setup()` runs once when the board starts or resets. It is normally used for initialization.

`loop()` runs repeatedly as long as the board is powered.

2. Functions We Used Before

`pinMode(pin, mode)`: configures a digital pin as INPUT or OUTPUT.

`digitalWrite(pin, value)`: writes HIGH or LOW to a digital output pin.

`digitalRead(pin)`: reads the current digital state of a pin and returns HIGH or LOW.

`delay(ms)`: pauses the program for the specified number of milliseconds.

3. Understanding pinMode()

Example:

```
pinMode(13, OUTPUT);
```

Here, 13 is the pin number and OUTPUT tells Arduino that the pin will be used to send a signal.

Common modes include INPUT and OUTPUT. In many button circuits, INPUT_PULLUP can also be used.

4. Understanding digitalWrite()

Example:

```
digitalWrite(13, HIGH);
```

This sends a HIGH signal to pin 13. For an LED connected appropriately, this can turn the LED on.

```
digitalWrite(13, LOW); turns the output off.
```

5. Understanding digitalRead()

Example:

```
int state = digitalRead(2);
```

The function reads pin 2 and stores the result in the variable state. A digital input is normally interpreted as HIGH or LOW.

This function is useful for buttons, switches, sensors, and other digital inputs.

6. Understanding delay()

Example:

```
delay(1000);
```

Arduino pauses for 1000 milliseconds, which equals 1 second.

Important idea: `delay()` blocks the program during the waiting period. This is simple and useful for introductory demonstrations.

7. What Is a Function?

A function is a named block of instructions designed to perform a specific task.

Functions help make programs easier to read, reuse, test, and maintain.

Arduino provides many ready-made functions, and programmers can also create their own functions.

8. Built-in vs User-defined Functions

Built-in functions: provided by the Arduino environment, such as `pinMode()`, `digitalWrite()`, `digitalRead()`, and `delay()`.

User-defined functions: created by the programmer for a particular task, such as `turnOnLED()` or `turnOffLED()`.

9. User-defined Function Example

Example:

```
void turnOn() {  
    digitalWrite(13, HIGH);  
}
```

Calling the function:

```
turnOn();
```

The function groups a repeated action under a meaningful name.

10. Demo 1 – Blinking LED

Code:

```
void setup() {  
    pinMode(13, OUTPUT);  
}  
  
void loop() {  
    digitalWrite(13, HIGH);  
    delay(1000);  
    digitalWrite(13, LOW);  
    delay(1000);  
}
```

Expected result: the LED turns on for one second and off for one second repeatedly.

11. Demo 2 – Button and LED

Code:

```
int button = 2;
int led = 13;

void setup() {
  pinMode(button, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  int state = digitalRead(button);

  if (state == HIGH) {
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

The key idea is: Input → Decision/Processing → Output.

12. Demo 3 – Traffic Light

Code:

```
int red = 13;
int yellow = 12;
int green = 11;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

void loop() {
  digitalWrite(green, HIGH);
  delay(3000);
  digitalWrite(green, LOW);

  digitalWrite(yellow, HIGH);
  delay(1000);
  digitalWrite(yellow, LOW);

  digitalWrite(red, HIGH);
  delay(3000);
  digitalWrite(red, LOW);
}
```

This demonstrates sequencing, digital outputs, and timing.

13. Code Reading: Predict Before Running

Ask students to predict what happens before running the program.

Example:

```
digitalWrite(13, HIGH);  
delay(2000);  
digitalWrite(13, LOW);
```

Question: How long does the LED stay on?

Answer: 2 seconds.

14. Common Mistakes

Using the wrong pin number.

Forgetting to configure the pin with pinMode().

Writing HIGH/LOW incorrectly.

Forgetting semicolons at the end of statements.

Confusing digitalWrite() with digitalWrite().

Using a button input without considering the required wiring/pull-up or pull-down arrangement.

15. Lecture Summary

setup() initializes the system; loop() repeatedly executes the main behavior.

pinMode() configures a pin.

digitalWrite() controls a digital output.

digitalRead() reads a digital input.

delay() creates a simple time delay.

Functions organize code into reusable tasks.

Good embedded programming connects inputs, processing, and outputs.

16. Questions for Students

1. What is the difference between setup() and loop()?
2. What does pinMode(13, OUTPUT) mean?
3. What is the difference between digitalWrite() and digitalWrite()?
4. What does delay(1000) do?
5. Why do programmers create user-defined functions?
6. Predict the output of a program that turns an LED HIGH, waits 500 ms, then turns it LOW.
7. In the button example, identify the input, processing step, and output.